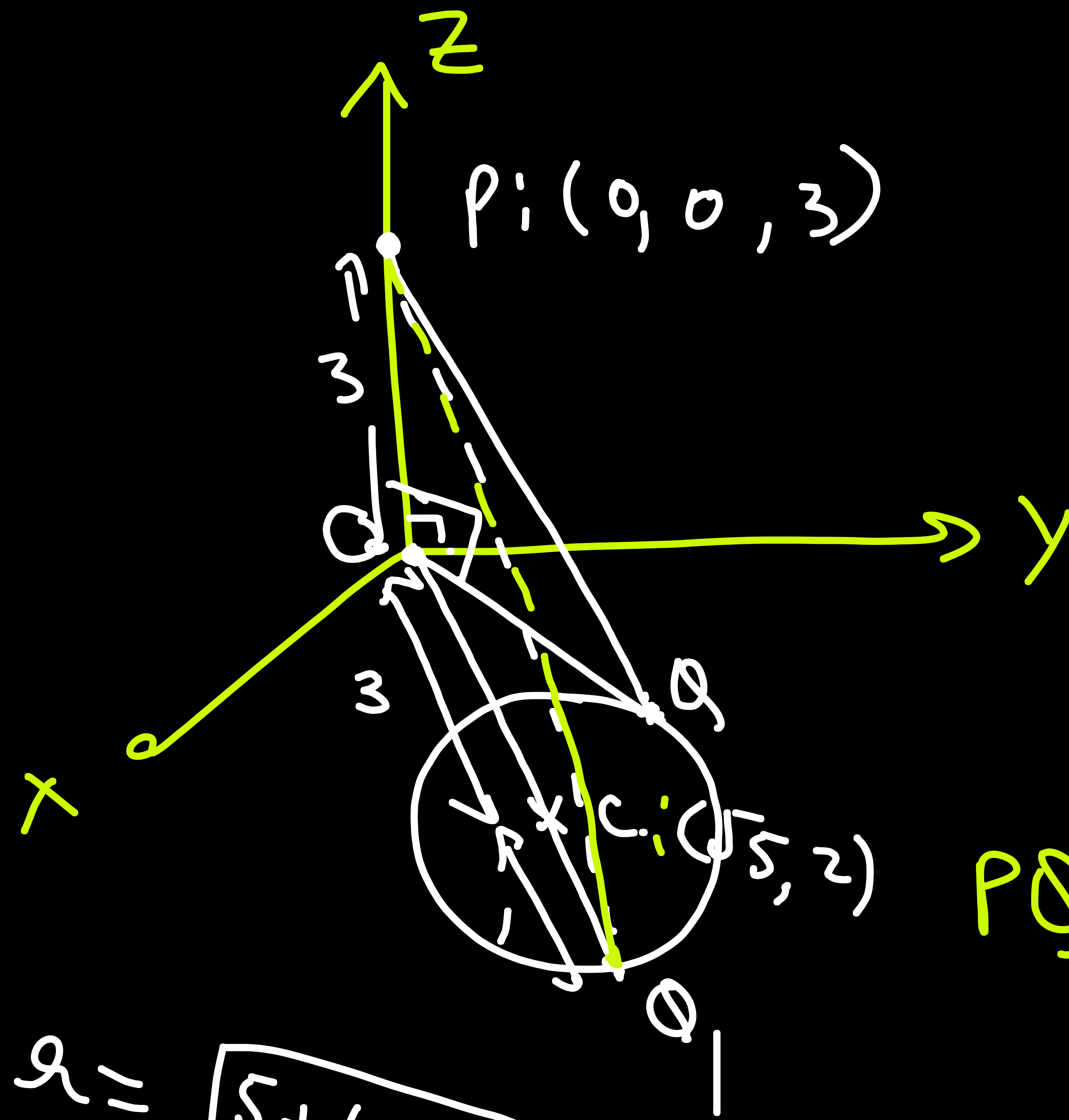


Que Find the maximum distance between the point $P: (0, 0, 3)$ and the circle $x^2 + y^2 - 2\sqrt{5}x - 4y + 8 = 0, z = 0$.

Sol



$$r = \sqrt{5 + 4 - 8} = 1$$

Let Q is any pt on circle and

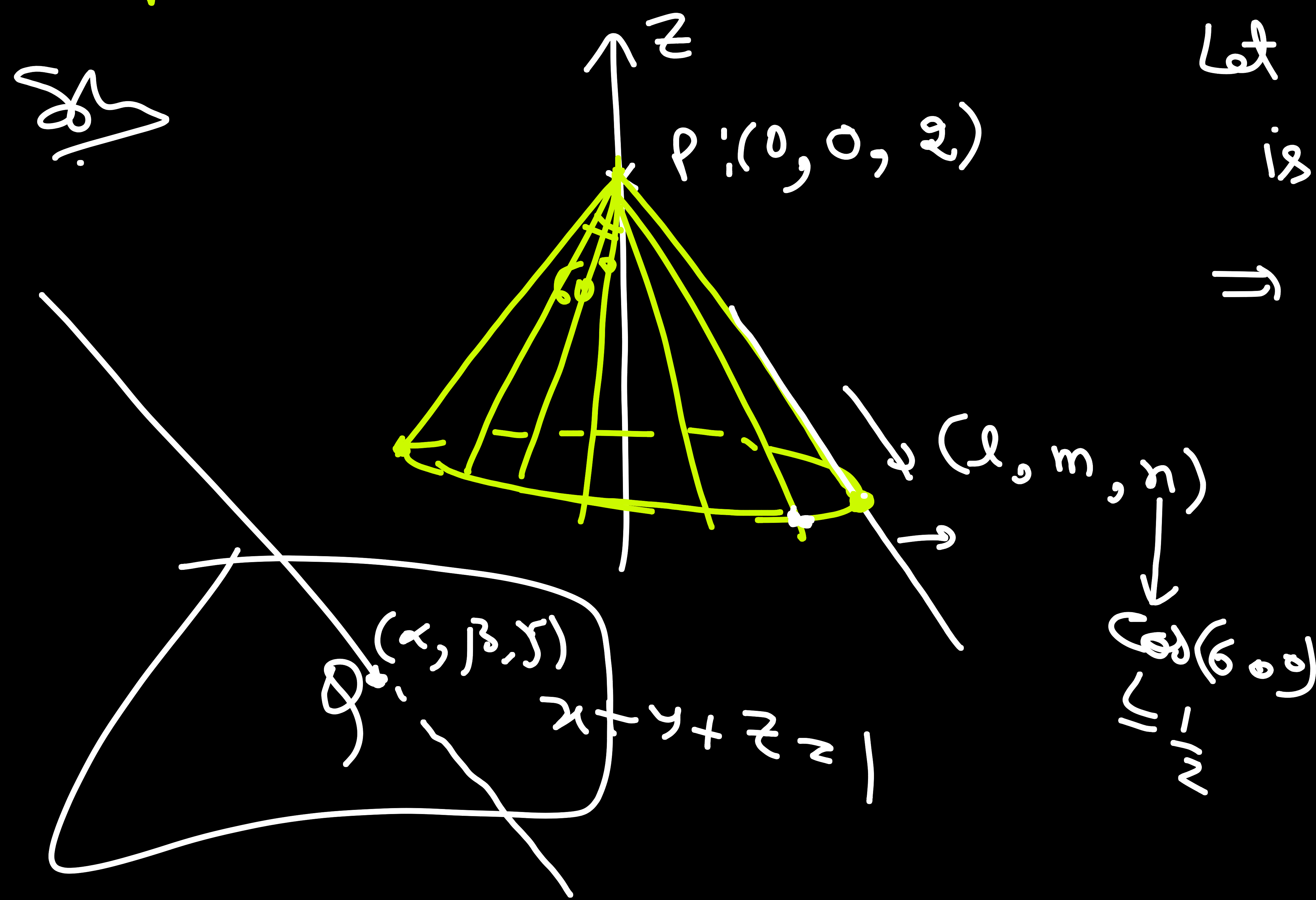
O is origin then

$$PQ^2 = \underbrace{OP^2}_{=9} + OQ^2$$

$\Rightarrow$ for PQ to be maximum, OQ should be maximum.

$$\begin{aligned} PQ)_{\max} &= \sqrt{OP^2 + OQ_1^2} \\ &= \sqrt{3^2 + 4^2} \\ &= \boxed{5} \end{aligned}$$

Ques A variable line passing through the point $P(0, 0, 2)$ always makes angle 60° with z -axis then find locus of point of intersection of this variable line and plane $x+y+z=1$.



Let direction cosine // to variable line is (l, m, n) $\odot$ $(l, m, \frac{1}{2})$

$\Rightarrow$ SLOPE of line is

$$L \equiv \frac{z-0}{l} = \frac{y-0}{m} = \frac{z-2}{\frac{1}{2}}$$

$\odot$ Let pt of X' of line L and given

Plane is $P(x, y, z)$

$$\frac{x}{l} = \frac{y}{m} = \frac{z-2}{\frac{1}{2}} = \lambda \Rightarrow \begin{aligned} x &= l\lambda \\ y &= m\lambda \\ z &= 2 + \frac{\lambda}{2} \end{aligned}$$

also $\alpha + \beta + \gamma = 1$

$$l\lambda + m\lambda + 2 + \frac{\lambda}{2} = 1$$

$$\lambda \left(l + m + \frac{1}{2} \right) = -1$$

$$\lambda = \frac{-1}{l + m + \frac{1}{2}}$$

$$l^2 + m^2 + \left(\frac{1}{2} \right)^2 = 1$$

$$\Rightarrow l^2 + m^2 = \frac{3}{4} \Rightarrow \frac{\alpha^2}{\lambda^2} + \frac{\beta^2}{\lambda^2} = \frac{3}{4}$$

$$\alpha^2 + \beta^2 = \frac{3}{4} \cdot \lambda^2$$

$$\Rightarrow \alpha^2 + \beta^2 = \frac{3}{4} \cdot (z-2)^2$$

$$\Rightarrow \boxed{x^2 + y^2 = 3(z-2)^2} \quad \text{Ans}$$

$$\alpha = l\lambda$$

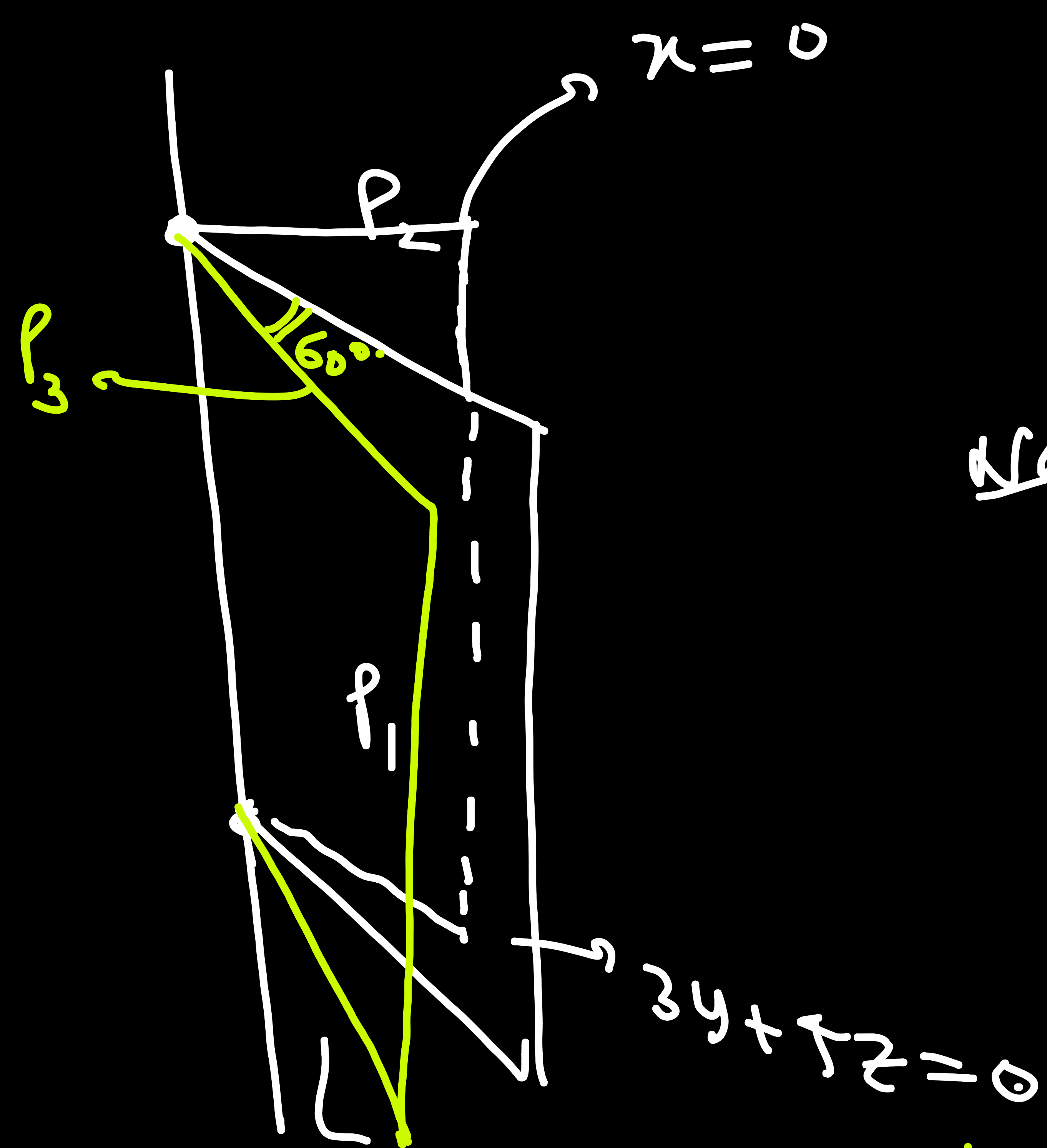
$$\beta = m\lambda$$

$$\gamma = 2 + \frac{\lambda}{2}$$

$$\Rightarrow 2(\gamma - 2) = \lambda$$

Ques The plane $3y+4z=0$ is rotated about its line of intersection with the plane $x=0$ through an angle 60° , then find its equation in new position.

Sol



Let required plane is $P_3 \equiv P_1 + \lambda P_2 = 0$

$$3y+4z + \lambda x = 0 \rightarrow \vec{n}_3 = (\lambda, 3, 4)$$

Now $\angle$ b/w P_1 & $P_3 = 60^\circ \Rightarrow \angle$ b/w $\vec{n}_1$ & $\vec{n}_3 = 60^\circ$.

$$\cos(60^\circ) = |\hat{n}_1 \cdot \hat{n}_2| \Rightarrow \pm \frac{1}{2} = \frac{\vec{n}_1 \cdot \vec{n}_2}{n_1 n_2}$$

$$\pm \frac{1}{2} = \frac{(0, 3, 4) \cdot (\lambda, 3, 4)}{5 \cdot \sqrt{\lambda^2 + 25}} = \frac{25}{5 \sqrt{\lambda^2 + 25}}$$

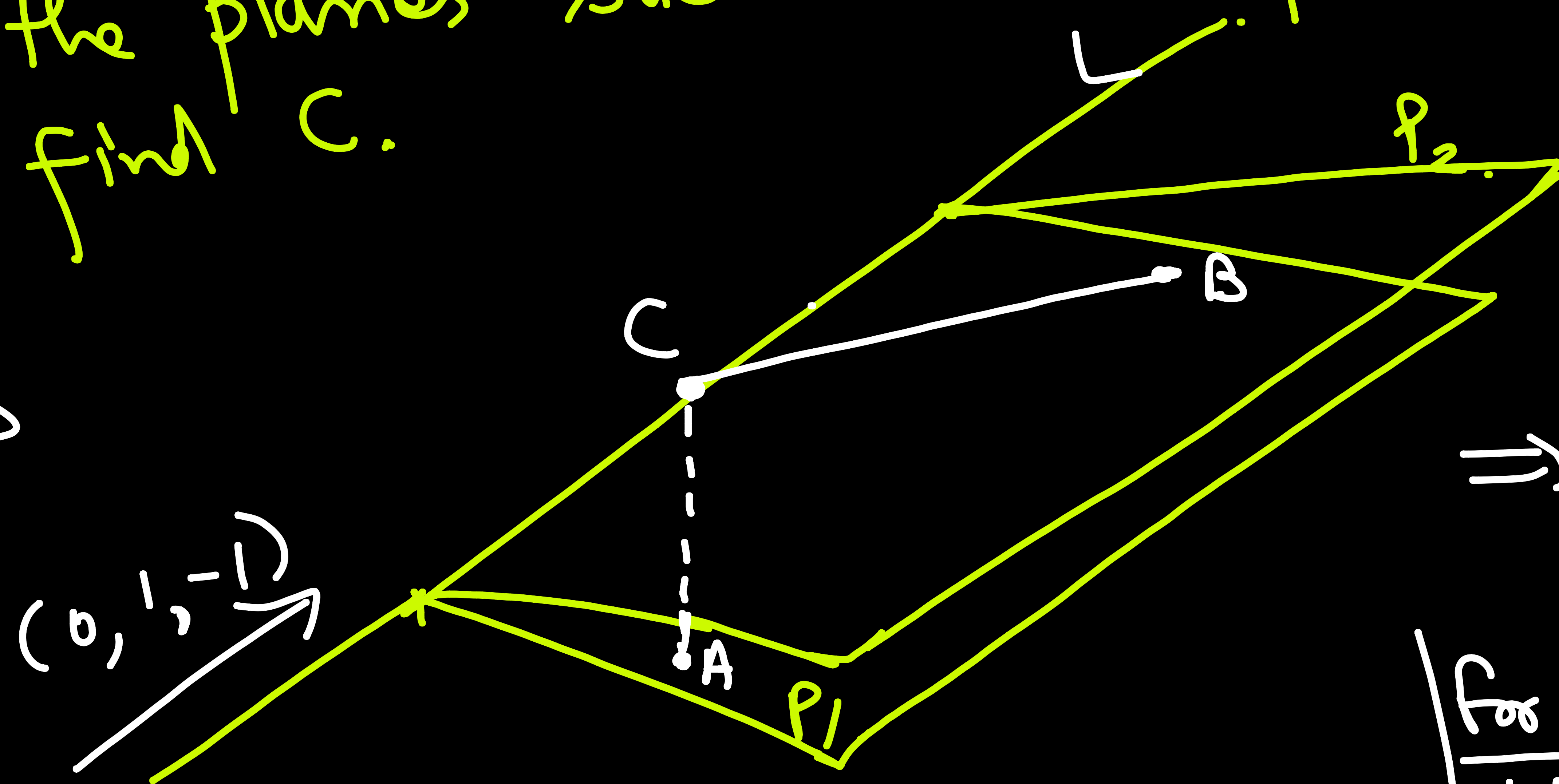
$$\lambda^2 + 25 = 100 \Rightarrow \lambda^2 = 75 \Rightarrow \lambda = \pm 5\sqrt{3}$$

$$\vec{n}_1 = (0, 3, 4)$$

$$P\text{-Plane} = \boxed{3y + 4z \pm 5\sqrt{3}x = 0}$$

Ques $A(3, -1, -1)$ and $B(3, 1, 0)$ lie on $x+y+z=1$ and $2x-y-z=5$ respectively. If C is a point lying on both the planes such that the perimeter of $\triangle ABC$ is minimum then find C .

Sol



$$P_1 \equiv x+y+z=1 \rightarrow \vec{n}_1 = (1, 1, 1)$$

$$P_2 \equiv 2x-y-z=5 \rightarrow \vec{n}_2 = (2, -1, -1)$$

$$\begin{aligned} \vec{n}_1 \times \vec{n}_2 &= (0, 3, -3) \\ &\equiv (0, 1, -1) \end{aligned}$$

Acc. to question we need to minimize $AB+BC+CA$

$\Rightarrow$ We need to minimize $AC+BC$
(as AB is const)

For pt on L .

$$\text{Let } z=0$$

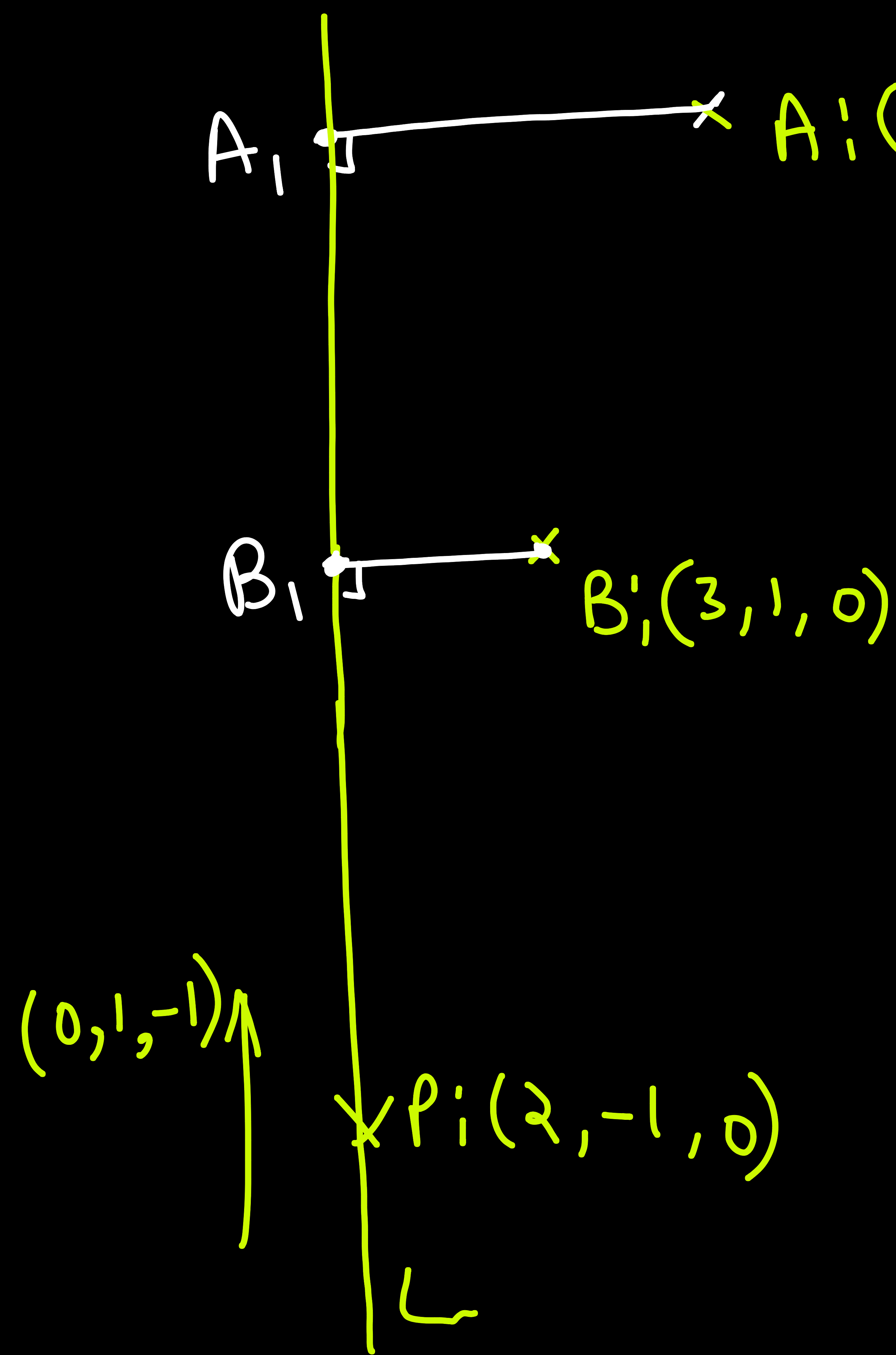
$$x+y=1$$

$$+2x-y=5$$

$$\Rightarrow x=2$$

$$\Rightarrow y=-1$$

$\Rightarrow$ A pt on L can be taken as $P_1(2, -1, 0)$.



$$L \equiv \frac{x-2}{0} = \frac{y+1}{1} = \frac{z}{-1} = \lambda$$

any pt on L can be taken as
 $(2, \lambda-1, -\lambda)$.

for A_1 $\overline{A_1 A} \cdot (0, 1, -1) = 0$. Let A_1 is $\rightarrow (2, \lambda-1, -\lambda)$
 $3, -1, -1$

$$(-1, \lambda, -\lambda+1) \cdot (0, 1, -1) = 0$$

$$\lambda + \lambda - 1 = 0 \Rightarrow \lambda = \frac{1}{2}$$

$$A_1: (2, -\frac{1}{2}, -\frac{1}{2})$$

for B_1 $\overline{B_1 B} \cdot (0, 1, -1) = 0$

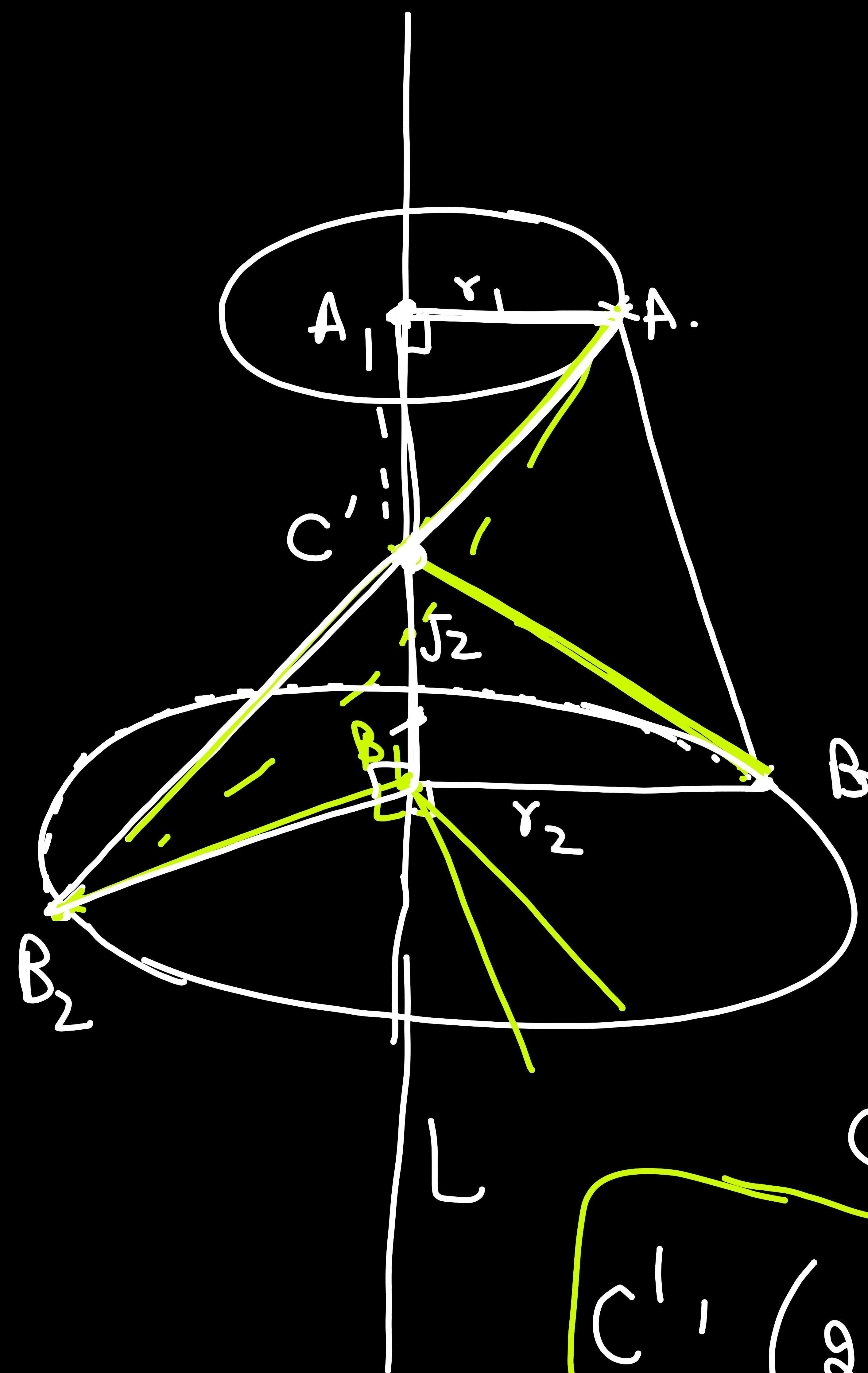
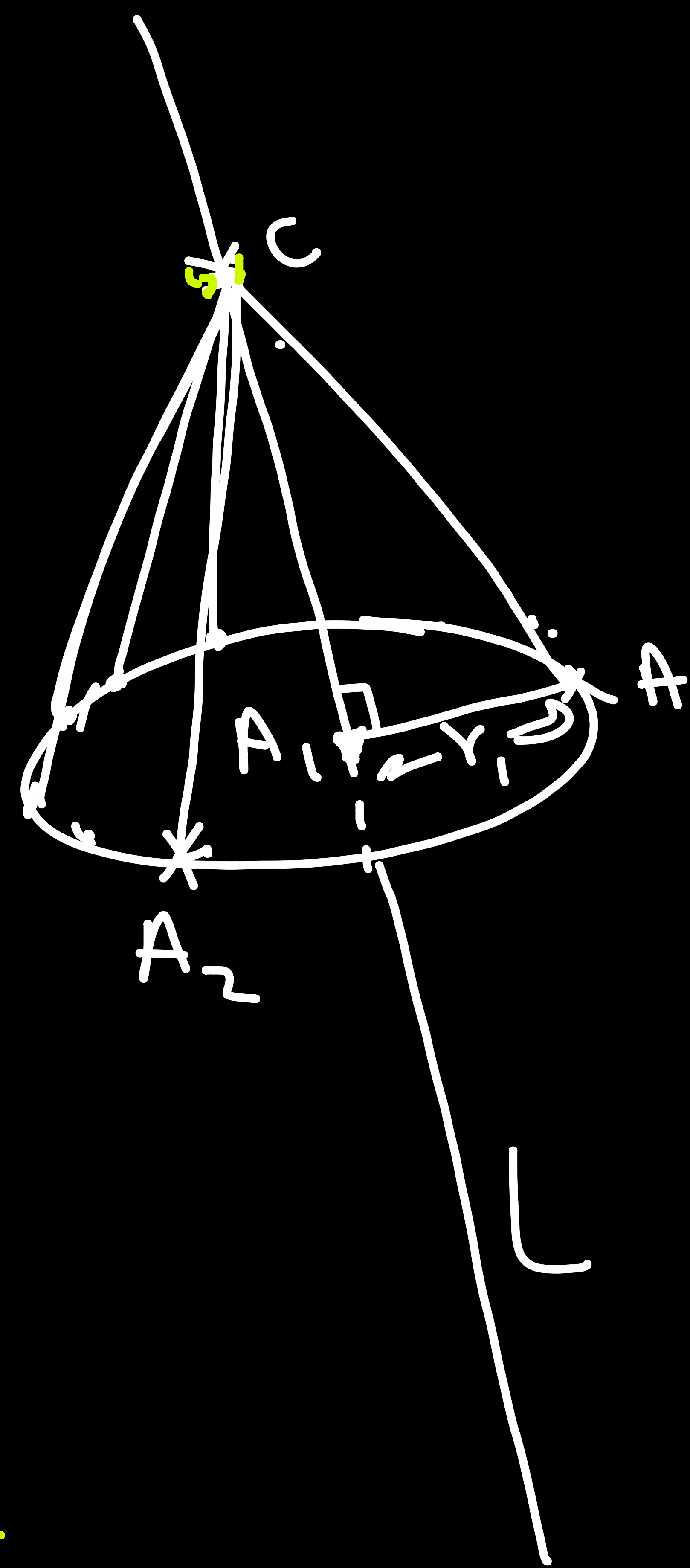
$$(-1, \lambda-2, -\lambda) \cdot (0, 1, -1) = 0$$

$$\lambda - 2 + \lambda = 0 \Rightarrow \lambda = 1$$

$$B_1: (2, 0, -1)$$

Let B_1 is
 $(2, \lambda-1, -\lambda)$
 $- 3, 1, 0$

 $(-1, \lambda-2, -\lambda)$



$$A: (3, -1, -1) \quad B = (3, 1, 0)$$

$$A_1: (2, -\frac{1}{2}, -\frac{1}{2}) \quad B_1 = (2, 0, -1)$$

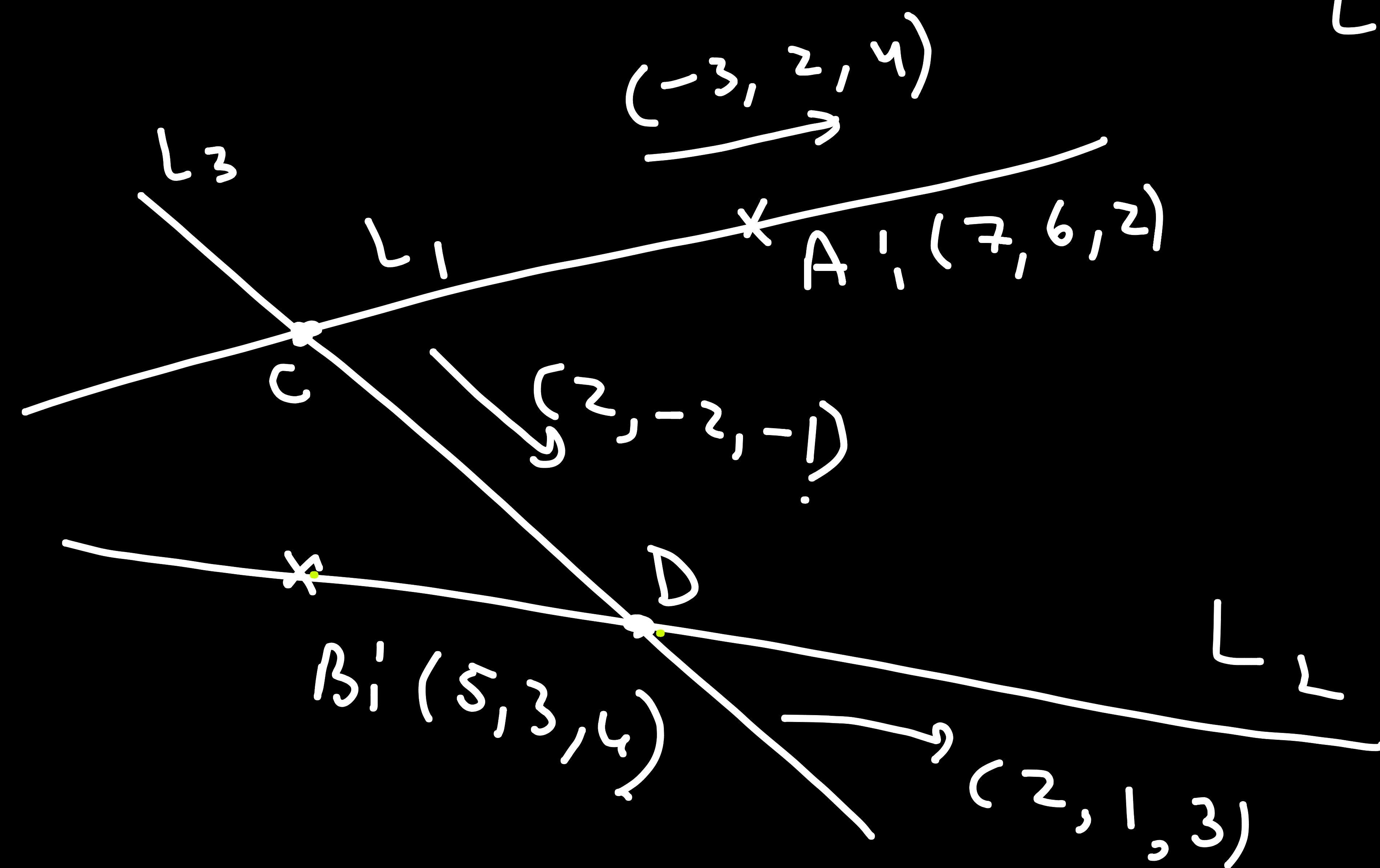
$$\left. \begin{aligned} r_1 &= A A_1 \\ &= \sqrt{1 + \frac{1}{4} + \frac{1}{4}} \\ &= \sqrt{\frac{3}{2}} \end{aligned} \right\} \begin{aligned} r_2 &= B B_1 \\ &= \sqrt{1 + 1 + 1} \\ &= \sqrt{3} \end{aligned}$$

$$\frac{A_1 C'}{B_1 C'} = \frac{x_1}{x_2} = \frac{\sqrt{3}}{\sqrt{2}} = \frac{1}{\sqrt{2}}.$$

$$C' : \left(\frac{2\sqrt{2}+1}{\sqrt{2}+1}, \frac{\sqrt{2}\left(-\frac{1}{2}\right)+1\cdot 0}{\sqrt{2}+1}, \frac{\sqrt{2}\left(-\frac{1}{2}\right)+1\cdot (-1)}{\sqrt{2}+1} \right)$$

$$C^1: \left(2, -\frac{1}{\sqrt{2}(\sqrt{2}+1)}, -\frac{1}{\sqrt{2}} \right) \quad \sqrt{2}+1$$

Que Let line L_1 has direction ratios $(-3, 2, 4)$ and passes through $A: (7, 6, 2)$ and line L_2 has d.r. $(2, 1, 3)$ and passes through $B: (5, 3, 4)$. If line L_3 has d.r. $(2, -2, -1)$ intersects L_1 & L_2 at C and D then find CD .



$$L_1 \equiv \frac{x-7}{-3} = \frac{y-6}{2} = \frac{z-2}{4} = \lambda$$

$$C: (7-3\lambda, 6+2\lambda, 2+4\lambda)$$

$$D: (5+2\mu, 3+\mu, 4+3\mu)$$

$$\overrightarrow{CD} = (-2+2\mu+3\lambda, -3+\mu-2\lambda, 2+3\mu-4\lambda)$$

$$\begin{aligned} -2+2\mu+3\lambda &= 3-\mu+2\lambda & \text{---} & \text{---} \\ 3\mu+\lambda &= 5 & \text{---} & \text{---} \end{aligned}$$

$$3\mu+\lambda=5$$

$$6\lambda-5\mu=7$$

$$6. \quad \begin{aligned} 3\mu + \lambda &= 5 \\ -5\mu + 6\lambda &= 7 \end{aligned}$$

$$\begin{aligned} 23\mu &= 23 \\ \mu &= 1 \end{aligned}$$

$$\lambda = 2.$$

$$\begin{aligned} \overrightarrow{CD} &= (-2 + 2\mu + 3\lambda, -3 + \mu - 2\lambda, 2 + 3\mu - 4\lambda) \\ &= (6, -6, -3) \end{aligned}$$

$$|\overrightarrow{CD}| = \sqrt{6^2 + 6^2 + 3^2}$$

$$= 3 \times 3 = \boxed{9}$$

Ques Find the points whose distances from $Y-Z$, $Z-X$ and $X-Y$ planes are in A.P. and whose distances from x , y , z axes are $\sqrt{13}$, $\sqrt{10}$ and $\sqrt{5}$ respectively.

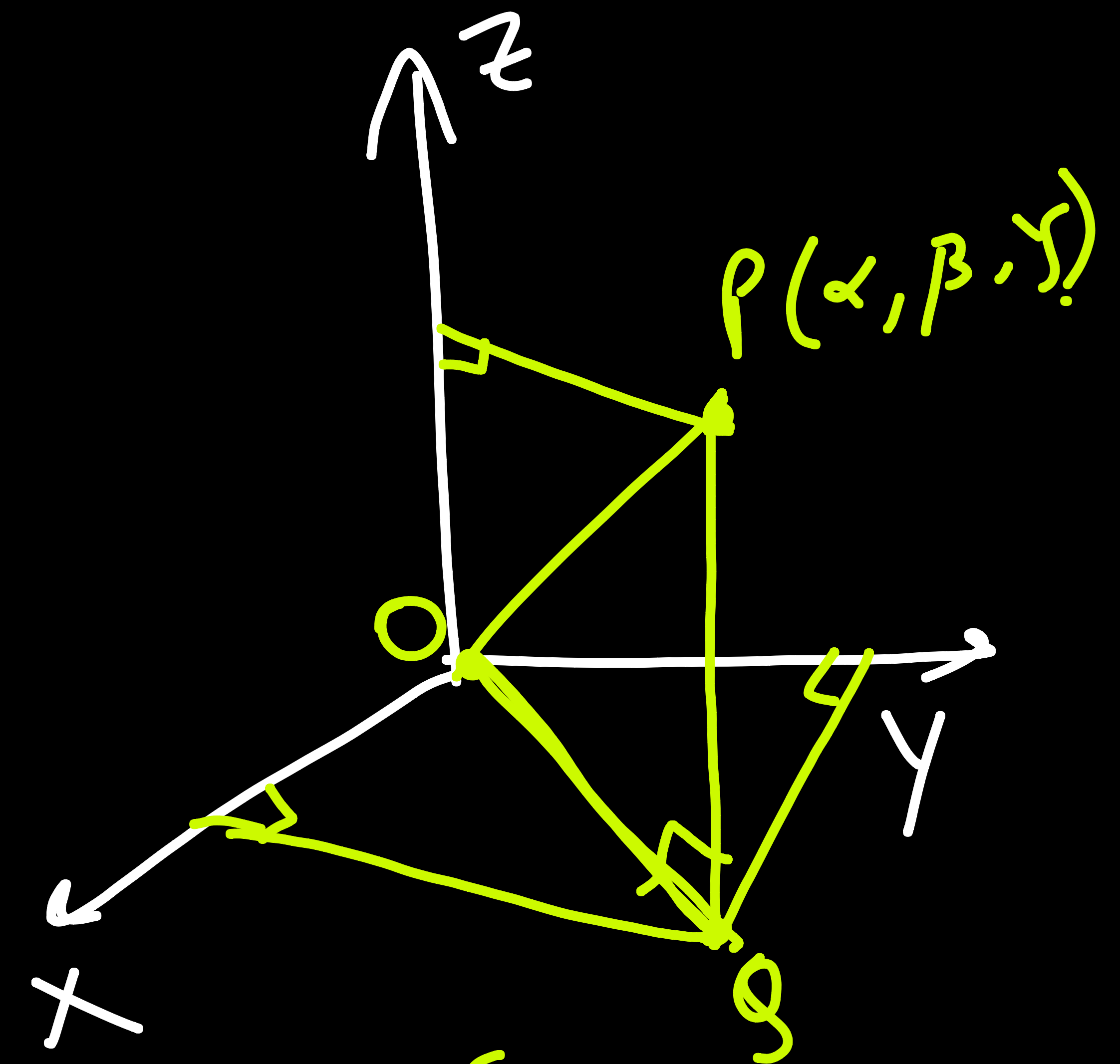
Sol Let distances are $a-d$, a , $a+d$
 then pt can be taken as $(\pm(a-d), \pm a, \pm(a+d))$

$$\begin{aligned}\sqrt{13} &= \sqrt{a^2 + (a+d)^2} \Rightarrow 2a^2 + 2ad + d^2 = 13 \\ \sqrt{10} &= \sqrt{(a-d)^2 + (a+d)^2} \Rightarrow 2a^2 + 2d^2 = 10 \\ \sqrt{5} &= \sqrt{(a-d)^2 + a^2} \Rightarrow 2a^2 - 2ad + d^2 = 5\end{aligned}$$

$$a^2 + d^2 = 5 \Rightarrow a^2 + \frac{4}{a^2} = 5$$

$$(a^2)^2 - 5a^2 + 4 = 0 \Rightarrow a^2 = 1, 4$$

$$a = \pm 1, \pm 2$$



distance of P from
 Z axis = OP
 $= \sqrt{OP^2 - OQ^2}$
 $= \sqrt{x^2 + y^2 + z^2 - y^2}$
 $= \sqrt{x^2 + z^2}$

$$ad = 2$$

~~$$a = \pm 1 \Rightarrow d = \pm 2$$~~

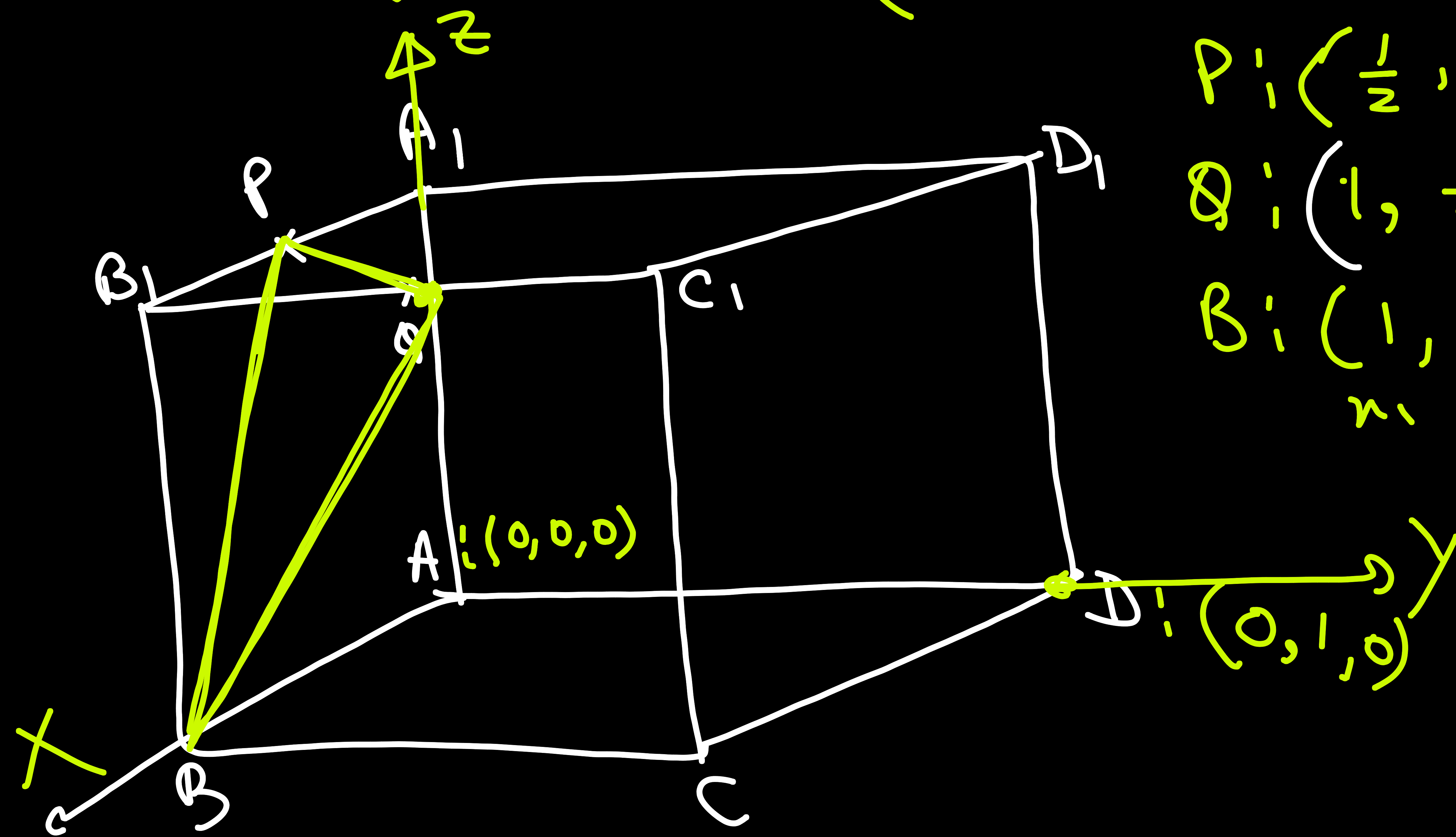
$$a = \pm 2 \Rightarrow d = \pm 1$$

$$(\pm 1, \pm 2, \pm 3)$$

$$\hookrightarrow 8 \text{ pts}$$

~~8~~

Ques $ABCD, A_1B_1C_1D_1$ is a cube of side length 1. If P, Q are the midpts of the edges B_1A_1 & B_1C_1 respectively then find distance of vertex D from PBQ . (given AA_1, BB_1, CC_1 & DD_1 are edges mutually $\perp$.)



$$\begin{aligned} P &: \left(\frac{1}{2}, 0, 1\right) \\ Q &: \left(1, \frac{1}{2}, 1\right) \\ B &: (1, 0, 0) \end{aligned} \quad \left\{ \begin{array}{l} \text{Eqn of Plane PBQ} \\ \begin{vmatrix} x-x_1 & y-y_1 & z-z_1 \\ x_2-x_1 & y_2-y_1 & z_2-z_1 \\ x_3-x_1 & y_3-y_1 & z_3-z_1 \end{vmatrix} = 0 \\ \begin{vmatrix} x-1 & y & z \\ 0 & 1/2 & 1 \\ -1/2 & 0 & 1 \end{vmatrix} = 0 \end{array} \right.$$

$$\Rightarrow \frac{1}{2}(x-1) - \frac{1}{2}y + \frac{1}{4}z = 0$$

$$2(x-1) - 2y + z = 0$$

$$2x - 2y + z - 2 = 0$$

$$2x - 2y + z - 2 = 0$$

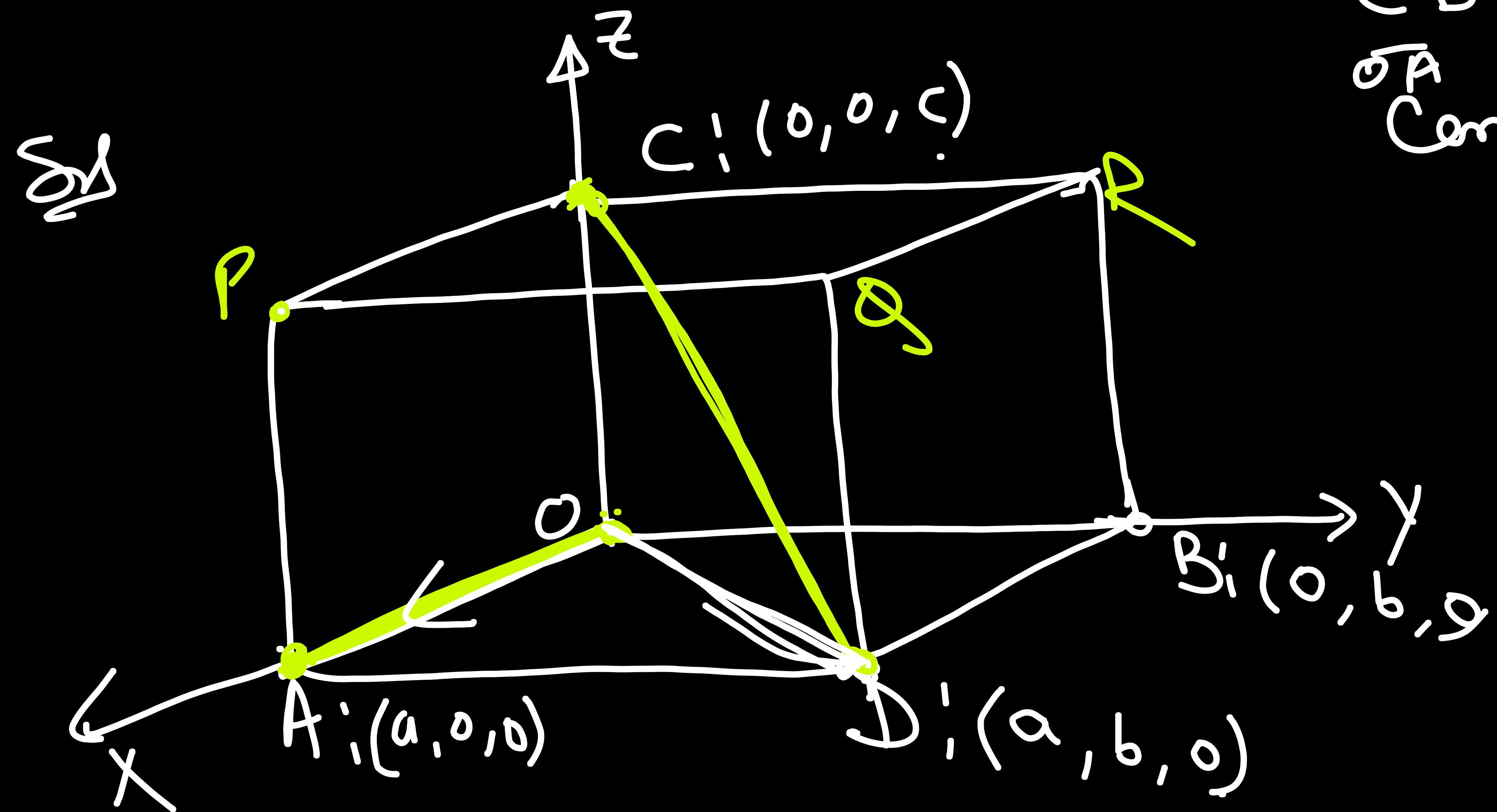
$$D: (0, 1, 0)$$

R. distance

$$\underline{\underline{L}} = \frac{|-2-2|}{\sqrt{4+4+1}} = \boxed{\frac{4}{3}} \rightarrow$$

Ques Let OA, OB, OC are coinitial edges of a cuboid and l, m, n be the shortest distances between the sides OA, OB, OC and their respective skew body diagonals to them respectively then find $\frac{\sum l^{-2}}{\sum OA^{-2}}$.

$$\begin{aligned}\overline{CD} &= (a, b, -c) \\ \overline{OA} &= (a, 0, 0) \\ \text{Common normal to } CD \& OA \text{ will be} \\ &= \overline{CD} \times \overline{OA} = (0, -ac, -ab) \\ \overline{n}_1 &= \left(0, \frac{1}{b}, \frac{1}{c}\right)\end{aligned}$$



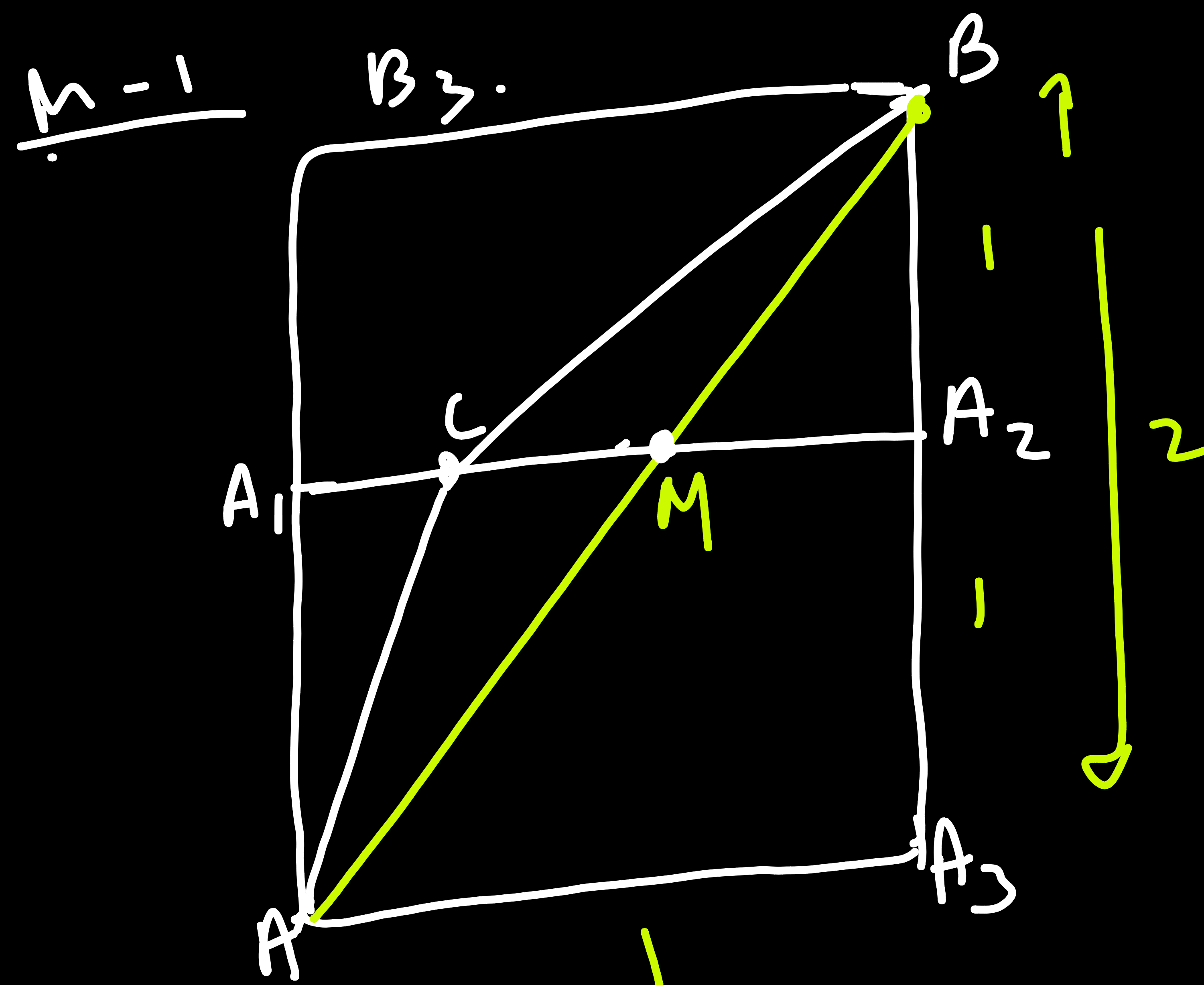
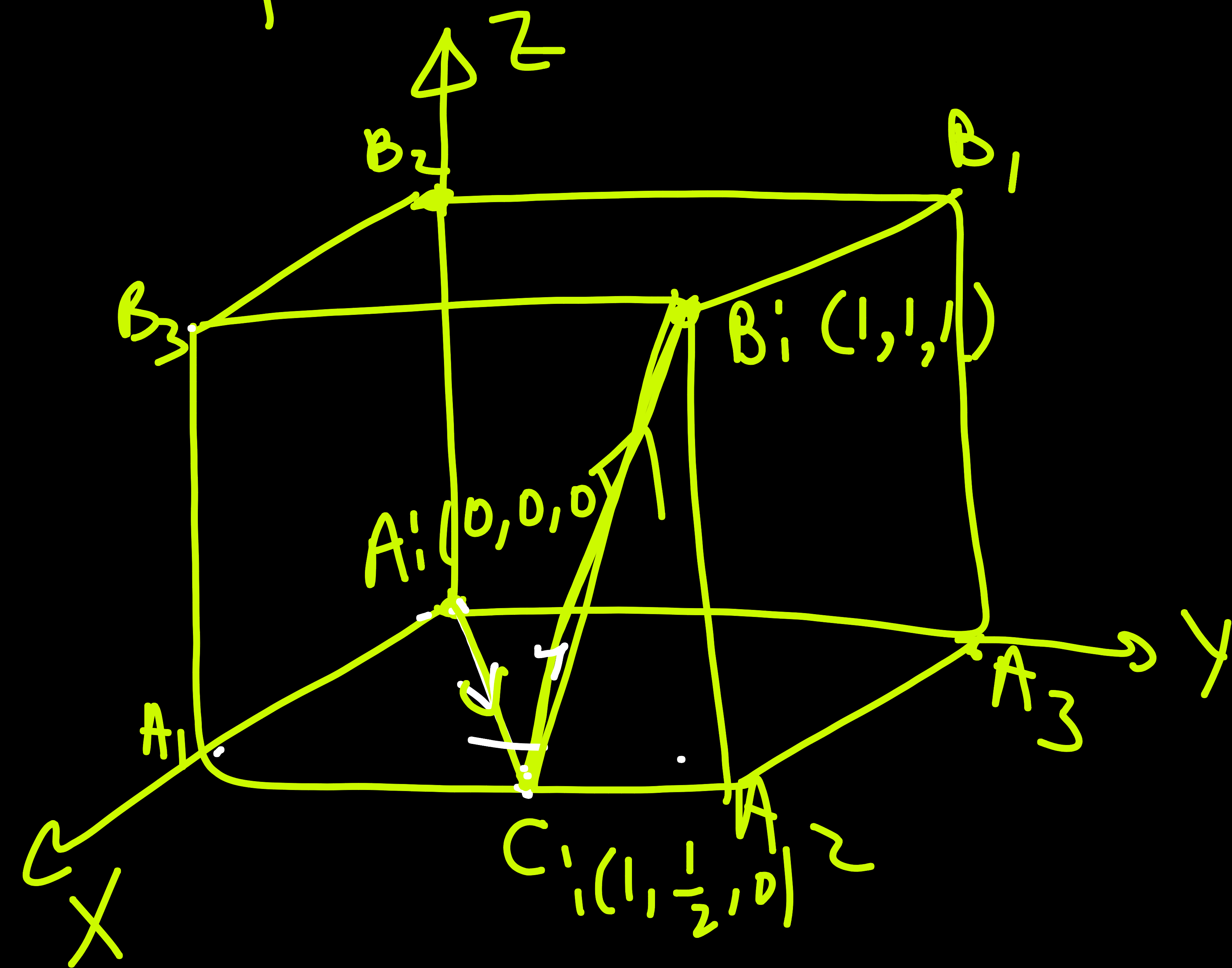
$$\begin{aligned}\text{Shortest distance b/w } OA \& CD \\ &= l = \text{Projection of } OD \text{ on } \overline{n}_1 \\ &= \frac{|\overline{OD} \cdot \overline{n}_1|}{|\overline{n}_1|} = \frac{|(a, b, 0) \cdot (0, \frac{1}{b}, \frac{1}{c})|}{\sqrt{\frac{1}{b^2} + \frac{1}{c^2}}} = \frac{1}{\sqrt{\frac{1}{b^2} + \frac{1}{c^2}}}\end{aligned}$$

$$\frac{1}{l^2} = \frac{1}{b^2} + \frac{1}{c^2}$$
$$\Rightarrow \sum \frac{1}{l^2} = \sum \frac{1}{b^2} + \frac{1}{c^2} = 2 \sum \frac{1}{a^2}$$

$$\Rightarrow \frac{\sum \frac{1}{l^2}}{\sum \frac{1}{a^2}} = \boxed{2.}$$

Q.E.D.

Que Let a cube has sides along $x-y-z$ axes and two of the opposite vertices are $A_1(0,0,0)$ and $B_1(1,1,1)$. An insect moves from A to B by crawling on the surface by traveling minimum distance. Find the possible direction cosines of the directions of its movements.



For min. distance
 C must be midpt of A_1A_2
 $\Rightarrow AC = (1, \frac{1}{2}, 0)$
 dir. cosine $= (\frac{2}{\sqrt{5}}, \frac{1}{\sqrt{5}}, 0)$
 $\overline{CB} = (0, \frac{1}{2}, 1)$
 dir cosine $= (0, \frac{1}{\sqrt{5}}, \frac{2}{\sqrt{5}})$

$$\sqrt{\frac{1}{4} + \frac{1}{4}} = \frac{\sqrt{2}}{2}$$

Que Let direction cosines (l, m, n) of two lines L_1 & L_2 satisfy $al + bm + cn = 0$ and $fmn + gnl + hlm = 0$. If L_1 & L_2 are mutually perpendicular then find $\sum \frac{f}{a}$.

$$al + bm + cn = 0 \xrightarrow{\div n} ax + by + c = 0$$

$$f(mn) + gnl + hlm = 0 \xrightarrow{\div n^2} fy + gx + hxy = 0.$$

$$-f\left(\frac{ax+c}{b}\right) + gx + h\left(-\frac{ax+c}{b}\right) = 0.$$

$\hookrightarrow x_1, x_2$

$$(l, m, n) \equiv \left(\frac{l}{n}, \frac{m}{n}, 1\right) \\ \equiv (x, y, 1)$$

$$\begin{array}{l} \text{Dir}_1 \swarrow \searrow \\ (x_1, y_1, 1) \quad (x_2, y_2, 1) \\ \text{Dir}_2 \end{array}$$

$$x_1 \cdot x_2 = \frac{-\frac{f \cdot c}{b}}{-\frac{a \cdot h}{b}} = \frac{f \cdot c}{a \cdot h}$$

$\parallel y$

$$y_1 y_2 = \frac{g \cdot c}{b \cdot h}$$

Consider $x_1 x_2 + y_1 y_2 + 1$

$x_1x_2 + y_1y_2 + 1 = 0$ if directions are mutually $\perp$ r

$$\frac{cf}{ah} + \frac{cg}{bh} + 1 = 0$$

•
also

$$\boxed{\frac{f}{a} + \frac{g}{b} + \frac{h}{c} = 0.}$$

Ques Let Line L_1 $x = ay + b$, $z = cy + d$ and L_2 $a'y + b' = x$ and $z = c'y + d'$ are perpendicular to each other then find the required condition.

Sol

$$\begin{array}{lcl}
 L_1 \rightarrow \textcircled{1} \begin{array}{l} x = ay + b \\ z = cy + d \end{array} & & L_2 \rightarrow \begin{array}{l} x = a'y + b' \\ \textcircled{2} z = c'y + d' \end{array} \\
 \textcircled{1} & & \textcircled{2} \\
 \frac{x-b}{a} = \frac{y}{1} = \frac{z-d}{c} & & \frac{x-b'}{a'} = \frac{y}{1} = \frac{z-d'}{c'}
 \end{array}$$

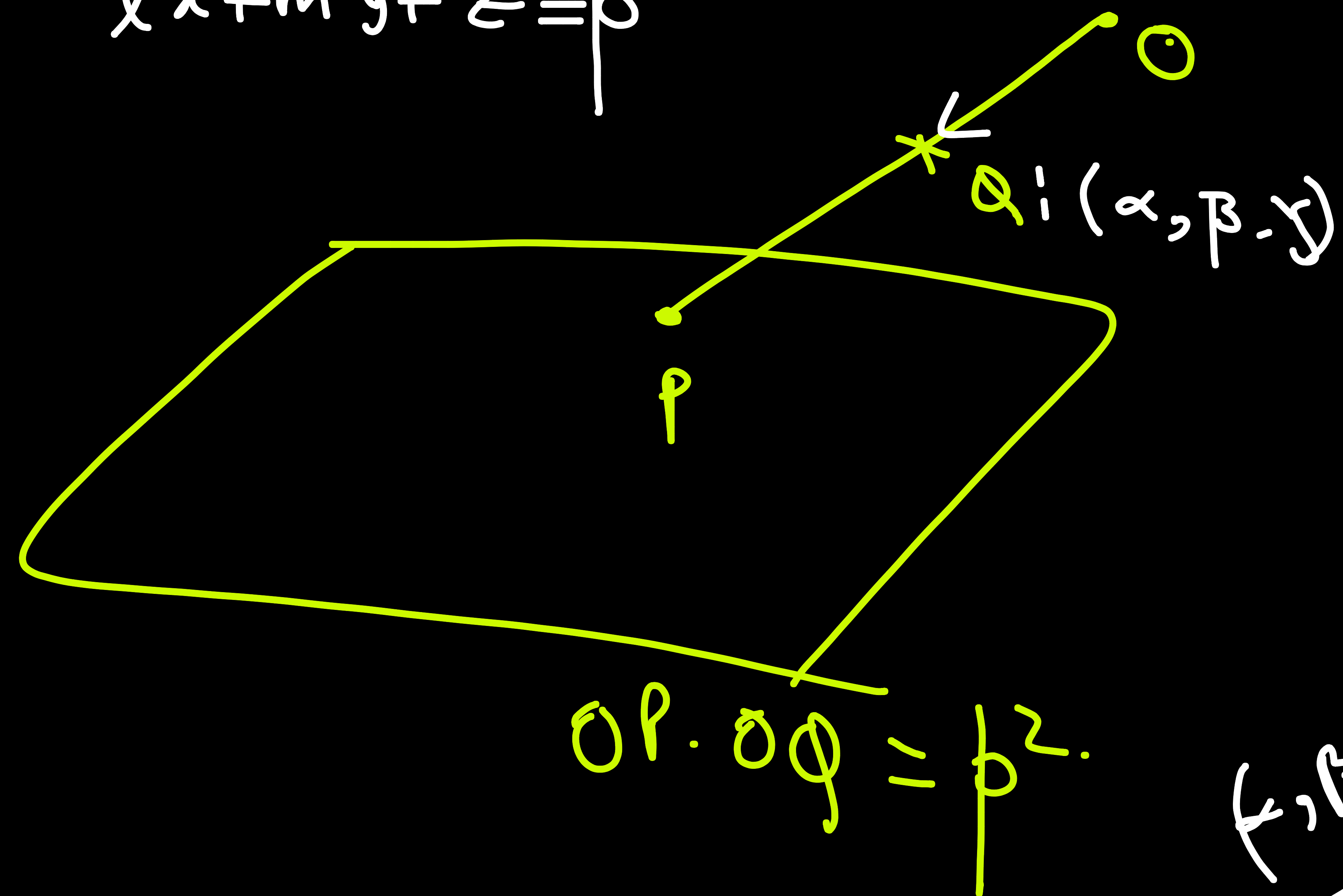
$$(a, 1, c) \cdot (a', 1, c') = 0.$$

$$\Rightarrow \boxed{aa' + cc' + 1 = 0} \quad \text{Ans}$$

Ques Let P be any point on the plane $lx + my + z = p$ and Q be a point on line OP such that $OP \cdot OQ = p^2$ then find the locus of Q .

Sol

$$lx + my + z = p$$



$$\vec{OP} = \frac{\vec{OQ}}{OQ} \cdot OP = \frac{p^2}{OQ^2} \cdot \vec{OQ}$$

$$= \left(\frac{p^2}{\alpha^2 + \beta^2 + \gamma^2} \right) (\alpha, \beta, \gamma)$$

$\vec{OP}$ will satisfy $lx + my + z = p$.

$$l \left(\frac{\alpha p^2}{\alpha^2 + \beta^2 + \gamma^2} \right) + m \left(\frac{\beta p^2}{\alpha^2 + \beta^2 + \gamma^2} \right) + \frac{\gamma p^2}{\alpha^2 + \beta^2 + \gamma^2} = p$$

(x, y, z)
 (α, β, γ)

$$lp\alpha + m\beta + pz = \alpha^2 + \beta^2 + \gamma^2$$